TrayGuard — Surgical Tray Training

EC ENGR 180DW — Final Design Review

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The Problem

- **9 defects per 100** surgical cases
- **55%** occur during tray assembly
- **10 min** average delay per affected case
- **$6.75–9.42M** estimated annual cost per hospital
- **44%** of packaging errors: wrong specification

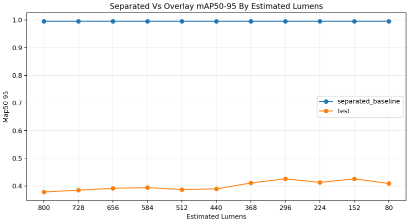
> *"The total count was correct. The specific instrument was wrong."*

Kelly and Crile forceps: Both have transverse serrations. Used for medium to large vessels. Crile's serrations along entire length of jaws. Kelly's serrations only on ½ of the distal jaws

Kelly vs Crile — same family, different serration pattern

Why Cameras Failed

- Seven staged experiments tested lighting, background, and tray layout
- The camera worked best only in clean lab conditions
- When instruments touched, detection quality dropped sharply
- The model learned **positions, not shapes**
- Each hospital would need its own labeled image dataset



mAP50–95 degradation under occlusion

The camera failed under the conditions trays actually create.

TrayGuard

- Training platform: **laptop + printer + cards**
- Camera scans which cards are on the table
- Scoring engine classifies every selection into **6 error categories**
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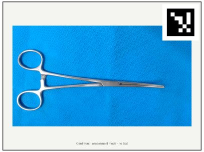
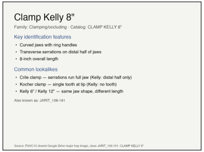
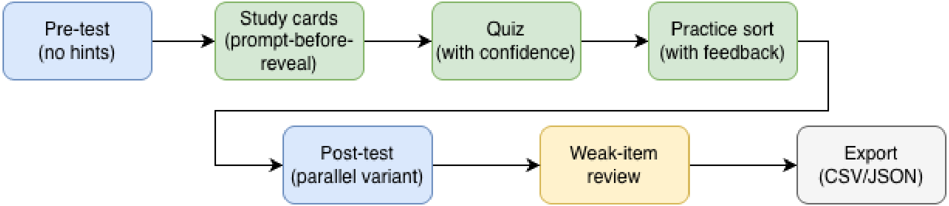




The real bottleneck is training the people who build trays.

How It Works



Pre / post use parallel variants. Study cards are retrieval practice. Practice gives error-specific feedback.

Pre / Post Test

Parallel variants

Different photo views

No hints

Baseline + measured gain

Study + Quiz

Retrieval-first cards

Prompt-before-reveal

Immediate feedback

Confidence capture

Practice Sort

Physical cards on table

Scored against count sheet

6 error categories

Weak-item review

What We Need

Prototype built. Study protocol designed.

**Looking for a hospital partner
to pilot the first real tray module.**

Open-source. Documented design handoff.

Contact the TrayGuard team through UCLA EC ENGR 180DW.

Direct material cost: **$53**. Requires: printer + existing laptop or tablet.

Evidence: Error by the Numbers

Alfred et al. (2021)

3,900 defects
41,799 cases

9 per 100

55% at assembly

Nichol et al. (2024)

236 errors
147 cases

10 min avg delay

$6.75–9.42M / yr

Zhu et al. (2019)

398 errors
33,839 packages

44% wrong spec

Largest error category

Three different methods converge on the same finding:

tray errors are recurring, measured, operational failures.

Three methods. One finding.

The Lookalike Problem: Kelly vs Crile

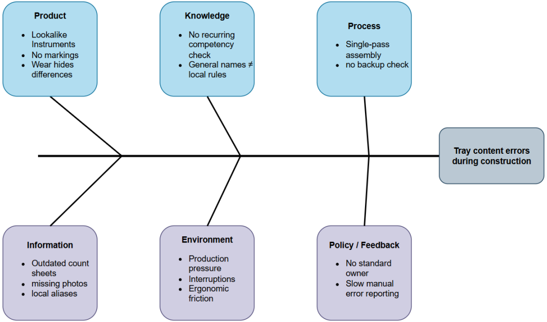
Kelly and Crile forceps: Both have transverse serrations. Used for medium to large vessels.
Crile's serrations along entire length of jaws. Kelly's serrations only on ½ of the distal jaws

Kelly (left) vs Crile (right) — serration pattern is the discriminating feature

- Same instrument family: both hemostatic clamps
- **Crile:** serrations cover the full jaw
- **Kelly:** serrations only cover distal half
- Sizes vary: 6", 7", 8"
- 44% of packaging errors are wrong specification

A few millimeters of serration — invisible at a glance.

Root Cause Analysis

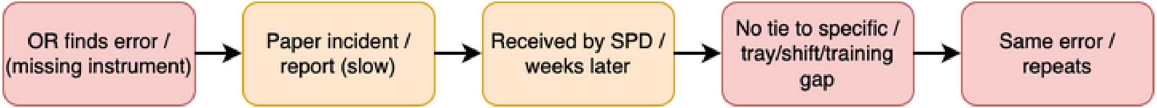


Tray errors come from instrument similarity, learner knowledge, process gaps, information flow, workspace conditions, and policy constraints.

Three recurring drivers: SPD as cost center, no licensure, no OR-to-SPD information system.

Tray errors come from a system, not a single mistake.

The Broken Feedback Loop



SPD workers often receive delayed or incomplete feedback after tray errors.

Evidence: 7 Staged CV Experiments

Experiment	Condition	Impact
Brightness (dark to bright)	80 to 800 lumens	Accuracy dropped
Brightness (bright to dark)	800 to 80 lumens	Same pattern
Background reflectivity	Matte vs reflective	Matte best, reflective worst
Clean layout	Instruments separated	Near-perfect detection
Overlap layout	Instruments touching	Detection dropped sharply
Real instruments	6 classes, fixed positions	Mixed class accuracy
Position check	Different data splits	Position effects appeared

The closer the setup got to a real tray, the worse detection became.

Evidence: Occlusion — Before and After

Separated — what you train on

Overlay — what a real tray looks like

You'd have to rearrange instruments to scan them.

Evidence: Position vs Shape Learning

- Accuracy changed by class depending on how the data was split
- Different instrument classes survived in different splits
- Only the **scalpel** transferred — the most visually distinct
- Each image had one of each class at a **fixed position** per session
- Model learned **which class at which location**, not visual features
- When position changed (different tray), the model failed

The model memorized positions, not shapes.

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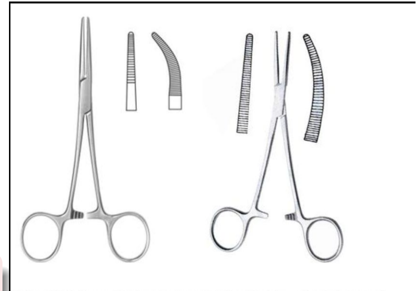

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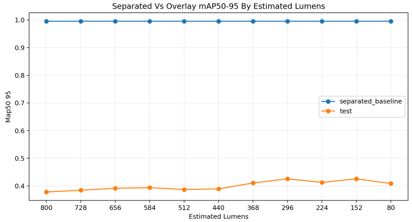

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The Pivot: TrayGuard

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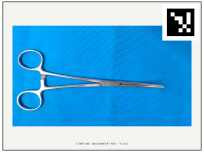
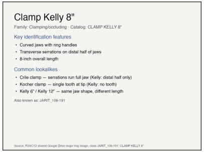




The real bottleneck is training the people who build trays.

Design: Error Categories

- **Correct** — item present, correct count
- **Missing** — required item not selected
- **Extra** — selected item not required
- **Wrong family** — selected item is the wrong type
- **Misidentified** — lookalike substitution
- **Wrong Count** — correct instrument, wrong quantity

Published evidence

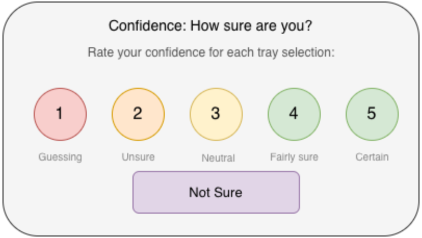
- **Alfred et al.** — missing, extra, wrong
- **Nichol et al.** — wrong, delays
- **Zhu et al.** — 44% misidentified

Each category maps to published literature.
When a learner improves, we know *which type* of error they reduced.

Same categories hospitals already track.

Design: Confidence Capture

- Confidence rated per item: **1–5 scale**
- Plus a **“Not sure”** flag
- **High-confidence error**: confidently wrong
- **Low-confidence correct**: fragile knowledge
- Both are flagged for review in CSV/JSON exports



Confidently wrong is not the same as fragile knowledge.

TrayGuard Scope

Current Scope

- Local tray familiarization
- Lookalike discrimination practice
- Simulated tray sort with error-category feedback
- Pre/post measurement with CSV/JSON export

Outside Current Scope

- CRCST certification
- Instrument damage or wear inspection
- Staffing and wage policy
- OR-to-SPD feedback infrastructure

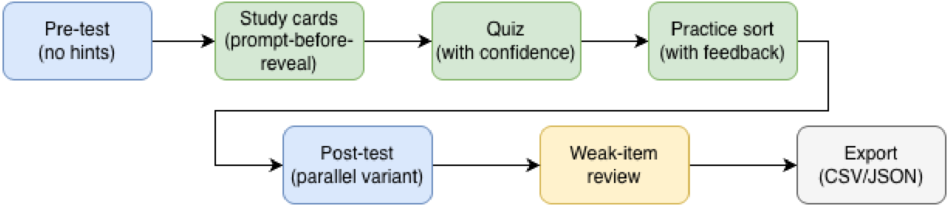
Claim bounded to: improvement on a simulated task in one session.

Why Training Was the Buildable Path

Approach	What it required
Automated tray checking (CV)	Per-hospital datasets, cross-manufacturer validation, regulatory review
Physical error-proofing (foam)	Requires instrument inventory, manufacturer relationships, per-tray tooling
Workflow redesign (dual check)	Requires policy authority, institutional buy-in, staffing changes
Decision-support kiosks	Requires deployment access, workflow integration, multi-stakeholder approval
Tray rationalization	Requires hospital utilization data, surgeon preferences, multi-department coordination
Competency tracking systems	Requires IT integration, HIPAA compliance, ongoing institutional commitment
Training	**Buildable with cards + laptop. Validatable with pre/post study. Creates durable local content any future solution needs.**

Training was the only buildable path.

Design: How It Works



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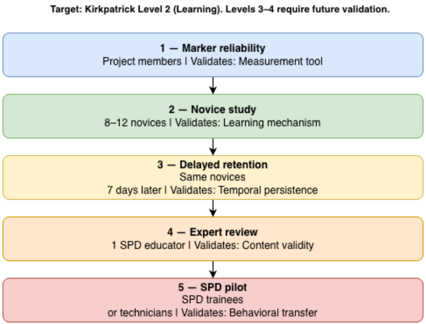
Evaluation: The Study Ladder



Study 0 Marker reliability — do the cards scan?
Study 1 Novice baseline — within-subject pre/post, 8–12 participants
Study 2 7-day retention — does improvement persist?
Study 3 Expert review — SPD educator validates content
Study 4 SPD workplace pilot — test with real technicians

Each step validates the next.

Evaluation: 7-Day Success Criteria

What counts as success after one week?

1. **Delayed accuracy retains at least half** of the immediate gain
2. **At least 5 percentage points above baseline**
3. **At least 70% of participants return** for the follow-up

If any criterion fails, the result supports **immediate learning only.**

The retention claim has a predefined threshold.

What We Accomplished

- **7 staged CV experiments** with documented negative result
- **Root cause analysis** across 6 categories + 3 structural drivers
- **Full system specification:** 7-step learning loop, scoring engine, confidence capture, CSV/JSON export
- **Study protocol** with explicit falsification criteria
- **Prototype web app** (FastAPI + vanilla JS) implementing the full learner loop
- **Durable tray module schema —** JSON files capturing local instrument knowledge

A rigorous no on CV is a real finding.

What We Need

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Direct material cost: **\$53**. Requires: printer + existing laptop or tablet.

Q&A: How Is This Different from Flashcards?

Feature	Flashcards	TrayGuard
Assessment	Recall only	Physical tray sort against count sheet
Feedback	Binary right/wrong	6 error categories mapped to hospital tracking
Context	Isolated instrument name	Distractors, lookalikes, required quantities
Confidence	None	1–5 scale + "Not sure" flag
Output	None	CSV/JSON export with hashed learner ID

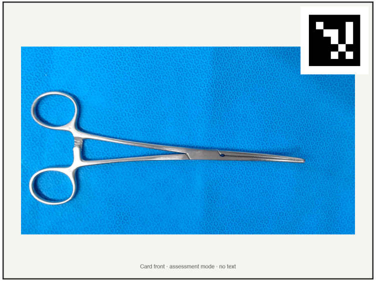
The applied task, not just memorization.

Q&A: Why AprilTags?

- **Faster detection** than QR codes — fewer bits to decode
- **Tolerates glare and partial occlusion** through built-in error correction
- **Stable 6-DOF pose estimation** handles camera angle variation
- **Manual fallback:** 3-digit ID on card back — type it if the camera misses
- Cards are re-printable paper. No real instruments needed.

Designed for tabletop cameras, not phones.

Q&A: Anti-Memorization Design



Assessment cards: neutral front — no name, no hint

- Pre-test and post-test use **matched parallel variants**
- Assessment cards show only the **AprilTag on the front**
- Pre and post use **different photo views** (A / B)
- Random seed ensures **unique layouts per session**

You cannot pass by memorizing card positions.

Q&A: Customizable Tray Module

Field	Example
Instrument ID	`hospital_kelly_6in`
Display name	Clamp Kelly 6"
Local aliases	KELLY 6", Kelly clamp 6
Required count	2
Lookalike pair	Kelly 6" vs Crile 5.5"
Feedback text	Check serration pattern

One tray module stores the local names, counts, and lookalike feedback.

- Single editable JSON file per hospital tray
- Instruments, local names, aliases, counts
- Distractors, lookalike pairs, features
- No code changes — edit file, reprint cards
- One module = **2–4 hours** SPD educator time
- Reusable for every new hire, competency check

Edit the file. Reprint the cards.

Q&A: Per-Hospital Data Burden

- Each hospital uses **different manufacturers, makes, and models**
- SPD lighting varies by **workstation, time of day, fixture condition**
- Tray layouts differ by **procedure type and surgeon preference**
- Cross-manufacturer validation needed — accuracy drops across brands
- Conservative estimate: **thousands of annotation hours per hospital**
- For ~**5,500** US hospitals: industry-scale infrastructure problem

Not a one-project problem.

Q&A: Cost & Next Steps

Prototype Bill of Materials

Item	Cost
Printed cards (15–16 instruments)	\$15–30
Tray surface / mat	\$5–10
Webcam (if laptop cam insufficient)	\$10–15
Total	**\$30–55**

Assumes existing laptop or tablet.

Priorities for the Next Team

1. Run novice baseline study
2. Expert review of tray content
3. 7-day retention check
4. SPD workplace pilot
5. Hospital partner for real module

\$53 in materials. Clinical validation first.

TrayGuard

Thank you. Questions?

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